Financial Time Series Analysis

Lecture 10

- TA Session: Friday, April 23, 11:00am-12:00pm
- TA Session: Monday, April 26, 7:30-8:30pm

Outline

- Review GARCH(1,1)
- Representing GARCH processes as ARMA processes
- Forecasting GARCH Models
- Example by R. Engle relevant to Risk Management
- Other GARCH models

From ARCH to GARCH

- The original ARCH model was created by R. Engle in 1982. It was generalized to GARCH (generalized autoregressive conditional heteroskedasticity) by Tim Bollerslev in 1986.
- The GARCH model allows the values of the X process to impact t he evolution of the conditional variance, σ_t.
- The GARCH(p,q) model is given by $\omega > 0,\ \alpha_i > 0, \beta_j > 0$:

$$a_t = \sigma_t \epsilon_t,$$

$$\{\epsilon_t\} \text{ is i.i.d.}, E(\epsilon_t) = 0, E(\epsilon_t^2) = 1, E(\epsilon_t^4) < \infty,$$

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i a_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2$$

GARCH(1,1):1

Consider a GARCH(1,1) model with $a_t = \sigma_t \epsilon_t$, and $\{\epsilon_t,\ t \geq 0\}$ (i.i.d. $E(\epsilon_t) = 0, E(\epsilon_t^2) = 1, E(\epsilon_t^4) < \infty$),

$$\sigma_t^2 = \omega + \alpha a_{t-1}^2 + \beta \sigma_{t-1}^2,$$

We will show $\{a_t^2\}$ is an ARMA(1,1) process.

$$\begin{aligned} E(a_t^2 | \mathcal{F}_{t-1}) &= E(\sigma_t^2 \epsilon_t^2 | \mathcal{F}_{t-1}) \\ &= \sigma_t^2 E(\epsilon_t^2 | \mathcal{F}_{t-1}) \\ &= \sigma_t^2 E(\epsilon_t^2) \\ &= \sigma_t^2 \cdot 1 \\ &= \sigma_t^2 \end{aligned}$$

GARCH(1,1):2

- Define $\eta_t = a_t^2 - \sigma_t^2$ or $a_t^2 = \eta_t + \sigma_t^2$
- $E(\eta_t|\mathcal{F}_{t-1}) = E(a_t^2 - \sigma_t^2|\mathcal{F}_{t-1}) = \sigma_t^2 - \sigma_t^2 = 0.$
- Consequently the $\{\eta_t,\ t \geq 0\}$ is a MDS and weak white noise, provided it has constant variance (homoscedastic).
- Some algebra gives

$$\begin{aligned}
\sigma_t^2 &= \omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\
&= \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2 \\
&= \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 (a_{t-1}^2 - \sigma_{t-1}^2) \\
&= \omega + (\alpha_1 + \beta_1) a_{t-1}^2 - \beta_1 \eta_{t-1}
\end{aligned}$$

GARCH(1,1):3

- Thus

$$a_t^2 = \sigma_t^2 + \eta_t = \omega + (\alpha_1 + \beta_1)a_{t-1}^2 - \beta_1\eta_{t-1} + \eta_t.$$

- Let $\nu = \omega/(1 - (\alpha_1 + \beta_1))$, so $\omega = \nu(1 - (\alpha_1 + \beta_1))$.
- Taking expectations across the equation and assuming stationarity, we find $E(a_t^2) = \nu$ and

$$a_t^2 = \nu(1 - (\alpha_1 + \beta_1)) + (\alpha_1 + \beta_1)(a_{t-1}^2) - \beta_1\eta_{t-1} + \eta_t,$$

or

$$a_t^2 - \nu = (\alpha_1 + \beta_1)(a_{t-1}^2 - \nu) - \beta_1\eta_{t-1} + \eta_t.$$

GARCH(1,1):4

- It follows that $\{a_t^2,\ t \geq 0\}$ is an ARMA(1,1) process with mean ν and AR(1) coefficient $\phi_1 = \alpha_1 + \beta_1$, and MA(1) coefficient $\theta_1 = -\beta_1$.
- Recall that for an ARMA(1,1), given by

$$X_t = \phi_0 + \phi_1 X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t$$

$$\rho_X(h) = \phi_1^{h-1} \frac{(\phi_1 + \theta_1)(1 + \phi_1 \theta_1)}{1 + 2\phi_1\theta_1 + \theta_1^2},$$

  so

- $$\rho_{a^2}(h) = (\alpha_1 + \beta_1)^{h-1} \frac{\alpha_1(1 - \alpha_1\beta_1 - \beta_1^2)}{1 - 2\alpha_1\beta_1 - \beta_1^2}$$

GARCH(p,q):1

- We have written the GARCH(1,1) as an ARMA(1,1) model with an MDS noise process. This can be generalized to the GARCH(p,q)

$$a_t = \sigma_t \epsilon_t,$$
$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i a_{t-i}^2 + \sum_{j=1}^{q} \beta_j \sigma_{t-j}^2,$$

where we assume $\{\epsilon_t\}$ are iid with mean 0 and variance 1.

- If we define $\alpha_i = 0$ if $i > p$ and $\beta_j = 0$ if $j > q$, then letting $\eta_t = a_t^2 - \sigma_t^2$ or $a_t^2 = \sigma_t^2 + \eta_t$, we find:

$$\begin{aligned} a_t^2 &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_i (a_{t-j}^2 - \sigma_{t-j}^2) \\ &= \omega + \sum_{i=1}^{\max(p,q)} (\alpha_i + \beta_i) a_{t-i}^2 + \eta_t - \sum_{j=1}^{\max(p,q)} \beta_j \eta_{t-j} \end{aligned}$$

GARCH(p,q):2

- η_t is a MDS, because $E(a_t^2|\mathcal{F}_{t-1}) = \sigma_t^2$, so $E(\eta_t|\mathcal{F}_{t-1}) = E(a_t^2|\mathcal{F}_{t-1}) - \sigma_{t-1}^2 = \sigma_{t-1}^2 - \sigma_{t-1}^2 = 0$.
- Consequently, $\{a_t^2\}$ is an ARMA(p,q) with AR coefficients $\alpha_i + \beta_i$ and MA coefficients $-\beta_i$.
- The variance of η_t is constant assuming $E(\epsilon_t^4) < \infty$ This can be done following the methods given in slides 7-9 in Lecture 9 given there for the ARCH(1) process. Consequently, η_t is an MDS and a white noise process.
- Because the error process is a white noise process, $\{a_t^2\}$ is an ARMA(p,q) process.

GARCH(p,q):3

- To ensure that the $\{a_t^2\}$ process is weakly stationary, we need the roots of the polynomial

$$1 - \sum_{j=1}^{\max(p,q)} (\alpha_i + \beta_i) z^i,$$

  to lie outside the unit circle.
- We usually assume that all of the α_i and β_j are positive, the weak stationary condition becomes

$$\sum_{i=1}^{p} \alpha_i + \sum_{j=1}^{q} \beta_j < 1.$$

- If this condition holds, then

$$E(a_t^2) = \omega / (1 - \sum_{i=1}^{p} \alpha_i - \sum_{j=1}^{q} \beta_j).$$

GARCH Forecasting: 1

- Suppose we wish to forecast X_{t+k} for some $k \geq 1$ where the $\{X_t\}$ process is ARMA(p_m, q_m) + GARCH(p_v, q_v).
- The GARCH(p_v, q_v) noise process is white noise, and we developed ARMA forecasting assuming white noise errors. So the point forecasts of X_{t+k} will be the same as if we assumed the errors were iid Gaussian.
- What is different is that the GARCH noise process will impact the variability of the forecast, specifically the forecast intervals.
- This is particularly important is risk management, as the concern is the quantile (e.g. 95% or 99%) of the loss distribution. The variance will impact the forecast of that quantile.

GARCH Forecasting: 2

- We showed that a GARCH(p_v, q_v) can be represented as an ARMA process with an MDS/white noise error process.
- Consequently, we can use the ARMA forecasting techniques on the GARCH model.
- We will first review ARMA forecasting and then apply it to GARCH forecasting.

GARCH Forecasting: 3

- We showed that the GARCH(1,1) can be written as an ARMA process with MDS noise, hence WN$(0, \sigma_a^2)$. Let us review how to forecast ARMA(p,q) processes:

$$X_t = \phi_0 + \sum_{i=1}^{p} \phi_i X_{t-i} + \sum_{j=1}^{q} \theta_j \epsilon_{t-j} + \epsilon_t.$$

- Recall that for forecasting a future value, say $\hat{X}_{t+k}$, we use the defining model above and forecast future values of the process, $(\hat{X}_{t+1}, \hat{X}_{t+2}, \cdots, \hat{X}_{t+k})$ sequentially.
- At each stage, we plug in the current data $(X_t, X_{t-1}, \cdots, X_{t-k})$ as called for by the model, and we plug in the estimates of the unknown parameters.

GARCH Forecasting: 4

- This leaves the ϵ_t terms in the equation. For current and previous values of ϵ_t, we plug in $(\hat{\epsilon}_t, \hat{\epsilon}_{t-1}, \cdots, \hat{\epsilon}_{t-q})$ going backward in time, and we plug in 0 going forward, i.e. $\hat{\epsilon}_{t+i} = 0$ for $i \geq 1$.
- For a GARCH(1,1) volatility process with parameters $\alpha_1 > 0$ and $\beta_1 > 0$ with $\alpha_1 + \beta_1 < 1$,

$$a_t^2 = \epsilon_t^2(\omega + (\alpha_1 + \beta_1)a_{t-1}^2 - \beta_1\eta_t).$$

- Let the constant unconditional variance be σ_a^2. Taking expectations across the equation, we find

$$\sigma_a^2 = \omega/(1 - \alpha_1 - \beta_1).$$

GARCH Forecasting: 5

- Suppose that the current time is t and we want to forecast $\hat{\sigma}^2_{t+k}$, the k-step ahead forecast of the volatility given $\mathcal{F}_t$
- Consider the one-step ahead forecast, $\hat{\sigma}^2_{t+1}$. We have

  $$\sigma^2_{t+1} = \omega + \alpha_1 a_t^2 + \beta_1 \sigma_t^2.$$

  The right-side is $\mathcal{F}_t$ measurable, so the above equation provides the one-step-ahead forecast, although estimates must be substituted for the unknown parameters, $\omega, \alpha_1, \beta_1$.
- For multi-step forecasts, $k \geq 1$, recall that $a_t^2 = \epsilon_t^2 \sigma_t^2$. So the volatility equation can be rewritten as

  $$\sigma^2_{t+1} = \omega + \alpha_1 \epsilon_t^2 \sigma_t^2 + \beta_1 \sigma_t^2$$

GARCH Forecasting: 6

- Or rearranging

$$\sigma_{t+1}^2 = \omega + (\alpha_1 + \beta_1)\sigma_t^2 + \alpha_1\sigma_t^2(\epsilon_t^2 - 1).$$

- The forecast at time $t+2$ is $E(\hat{\sigma}_{t+2}^2 | \mathcal{F}_t)$.
- Changing $t+1$ to $t+2$ in the above equation and taking $E(\cdot|\mathcal{F}_t)$ and noting $E(\epsilon_{t+1}^2 - 1|\mathcal{F}_t) = 0$, we find

$$\hat{\sigma}_{t+2}^2 = \omega + (\alpha_1 + \beta_1)\hat{\sigma}_{t+1}^2.$$

This generalizes to

GARCH Forecasting: 7

$$\hat{\sigma}^2_{t+k} = \omega + (\alpha_1 + \beta_1)\hat{\sigma}_{t+k-1}.$$

We let $\lambda = \alpha_1 + \beta_1$ and iterate backward:

$$\begin{aligned}
\hat{\sigma}^2_{t+k} &= \omega + \lambda\hat{\sigma}^2_{t+k-1} \\
&= \omega(1+\lambda) + \lambda^2\hat{\sigma}^2_{t+k-2} \\
&= \cdots \\
&= \omega(1 + \lambda + \cdots + \lambda^{k-2}) + \lambda^{k-1}\hat{\sigma}^2_{t+1}
\end{aligned}$$

As $k \to \infty$, assuming $\lambda < 1$

$$\hat{\sigma}^2_{t+k} \to \sigma^2 = \omega/(1-\lambda).$$

The half-life of the volatility difference is approximately $\lambda^T = \frac{1}{2}$, so $T \approx -\frac{\log 2}{\log \lambda} = -\frac{.693}{\log \lambda}$

Engle Test for ARCH Effects

IBM 1/1/2011 - 4/16/21

Fit by GARCH(1,1)

am = arch_model(logret)

res = am.fit(update_freq=5)

std_resid = res.resid/res.conditional_volatility

statsmodels.stats.diagnostic.het_arch(std_resid)

(7.865826004902203, 0.9999359555318335, Engle test

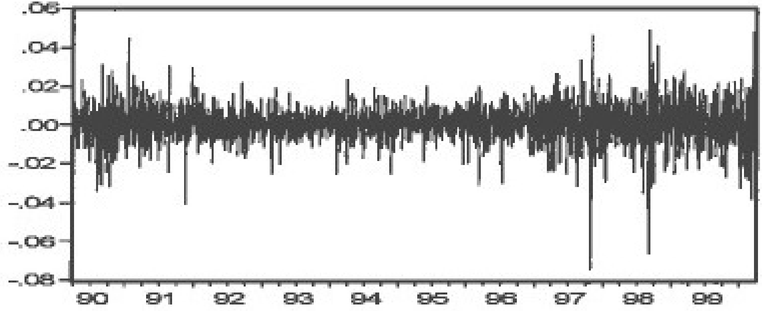
0.27859604468156574, 0.9999390381653693) F test

GARCH Forecasting Example

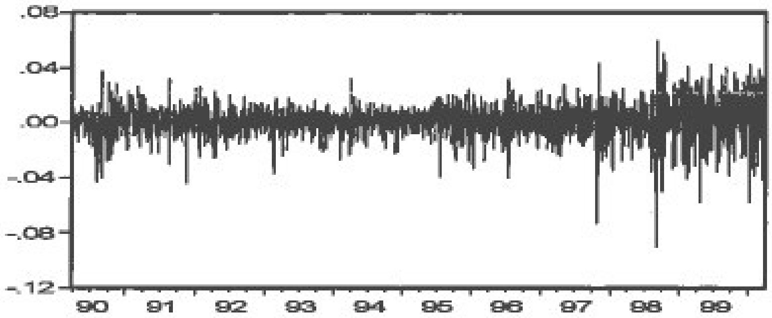
R. Engle presented an example of the use of GARCH models in Risk Management, ‘GARCH 101: The Use of ARCH/GARCH models in applied econometrics,” *Journal of Economic Perspectives*, **15(4)**, 2001, 157-168.

- He used a GARCH(1,1) model to estimate $\text{VaR}_{.99}$ for a \$1M portfolio on March 23, 2000.
- The portfolio consists of 50% NASDAQ, 30% Dow Jones and 20% long bonds (10 year constant maturity treasury bond). The proportion was kept constant through rebalancing.
- This date was chosen to be just before a large market slide and a time of high volatility.
- Consider the 10 year prior history:

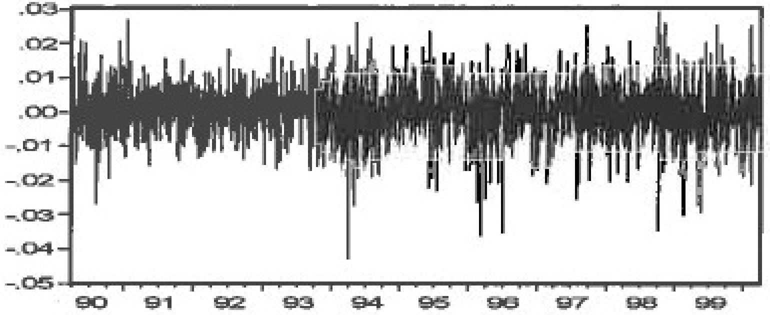
Dow Jones



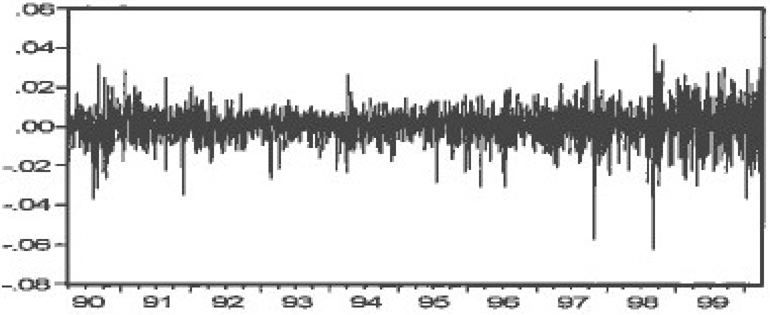
NASDAQ



Rates



Portfolio



Relevance to Risk Management: 1

3/23/1990 to 3/23/2000

	Dow Jones	NASDAQ	Rate	Portfolio
Mean	0.0005	0.0009	0.0001	0.0007
Std Dev	0.0090	0.0115	0.0073	0.0083
Skewness	-0.3593	-0.5310	-0.2031	-0.4738
Kurtosis	8.3288	7.4936	4.9579	7.0026

Relevance to Risk Management: 2

- If we consider a portfolio whose current value is $V(t)$. Suppose the portfolio is held constant over a time window of length Δ. The loss over this period is $L = V(t) - V(t + \Delta)$
- L is a random variable whose value depends upon Δ and the change in risk factors over the interval (stock prices, rates, commodity prices, etc.).
- A risk measure is some “feature” of this random variable, like its standard deviation.
- VaR = value at risk is a particular quantile of the loss distribution.
- Shortfall risk is $E(L|L > Var_{\alpha})$

Relevance to Risk Management: 3

- The VaR value can be computed by a number of methods:
  - Historical data - look at the last *n* values of the portfolio. Use the α quantile of the empirical distribution to estimate the VaR value.
  - Model the loss distribution and compute the quantile from the model.
  - Forecast, say using GARCH, to generate a point prediction for the portfolio and a standard deviation for the time of the forecast.. Use these two quantities to generate a VaR (quantile) estimate.

Relevance to Risk Management: 4

"I now turn to the question of how the econometrician can possibly estimate an equation like the GARCH(1,1) equation:

$$X_t = m_t + \epsilon_t \sigma_t,$$

$$\sigma_{t+1}^2 = \omega + \alpha(X_t - m_t)^2 + \beta\sigma_t^2.$$

when the only variable on which there are data is $\{X_t\}$, the returns or log-returns. The simple answer is to use maximum likelihood by substituting σ_t^2 in the normal likelihood and then maximizing with respect to the parameters. Many software packages will facilitate this automatically."

Relevance to Risk Management: 5

The values of the three parameters in the GARCH(1,1) model are

- $\omega = 1.4E - 06(.0018)$
- $\alpha_1 = .0772(.0000)$
- $\beta_1 = .9046(.0000)$

"Consider a trader who knows that the long-run average daily standard deviation of the S&P 500 is 1 percent, that the forecast made yesterday was 2 percent and the unexpected return observed today is 3 percent. Obviously, this is a high volatility period, and todfay is especially volatile, which suggests that the forecast for tomorrow could be even higher. However, the fact that the long-term average is only 1 percent might lead the forecaster to lower the forecast."

ACF of m_t

Table 2

Autocorrelations of Squared Portfolio Returns

	AC	*Q-Stat*	*Prob*
1	0.210	115.07	0.000
2	0.183	202.64	0.000
3	0.116	237.59	0.000
4	0.082	255.13	0.000
5	0.122	294.11	0.000
6	0.163	363.85	0.000
7	0.090	384.95	0.000
8	0.099	410.77	0.000
9	0.081	427.88	0.000
10	0.081	445.03	0.000
11	0.069	457.68	0.000
12	0.080	474.29	0.000
13	0.076	489.42	0.000
14	0.074	503.99	0.000
15	0.083	521.98	0.000

Sample: March 23, 1990 to March 23, 2000.

ACF of a_t^2 model

Table 4

Autocorrelations of Squared Standardized Residuals

	AC	*Q-Stat*	*Prob*
1	0.005	0.0589	0.808
2	0.039	4.0240	0.134
3	−0.011	4.3367	0.227
4	−0.017	5.0981	0.277
5	0.002	5.1046	0.403
6	0.009	5.3228	0.503
7	−0.015	5.8836	0.553
8	−0.013	6.3272	0.611
9	−0.024	7.8169	0.553
10	−0.006	7.9043	0.638
11	−0.023	9.3163	0.593
12	−0.013	9.7897	0.634
13	−0.003	9.8110	0.709
14	0.009	10.038	0.759
15	−0.012	10.444	0.791

Relevance to Risk Management: 6

“The best strategy depends upon the dependence between days. If these 3 numbers are each squared and weighted equally, then the new forecast would the

$$\sqrt{(1+4+9)/3} = 2.16.$$

However, rather than weighting these equally, it is generally found for daily data that weights such as those in th eempirical example (.02, .9, .08) are much more accurate. Hence the forecast is

$$\sqrt{.02 * 1 + .9 * 4 + .08 * 9} = 2.08.$$

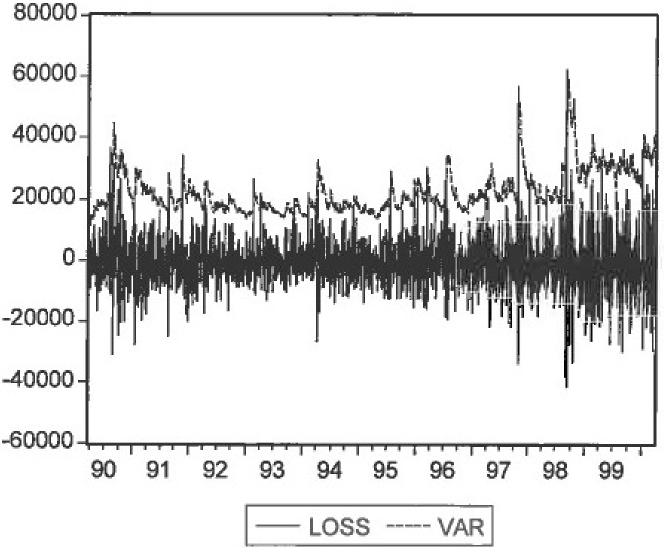
Relevance to Risk Management: 7

- The $\text{VaR}_{.99}$ values for the portfolio calculated from historical data:
  - \$22,477 using the 10-year sample
  - \$24,653 using the most recent year
  - \$35,159 using 1/1/2000 to 3/23/2000
- Fitting a GARCH(1,1) model: parameters estimates and p-value:
  - $\omega = 1.4E - 06(.0018)$
  - $\alpha_1 = .0772(.0000)$
  - $\beta_1 = .9046(.0000)$

Relevance to Risk Management: 8

- Good model fit and the estimated coefficients sum to less than 1, but are close to 1, so we have slow mean reversion.
- The forecast next day standard deviation is 0.0146, which is much larger than .0083, the 10 year average standard deviation. So this day is in a period of high volatility.
- The residuals are relatively heavy tailed, so multiply by 2.844 (instead of 2.327 for a standard normal distribution).
- $VaR_{.99}$ value is $39,996.

Portfolio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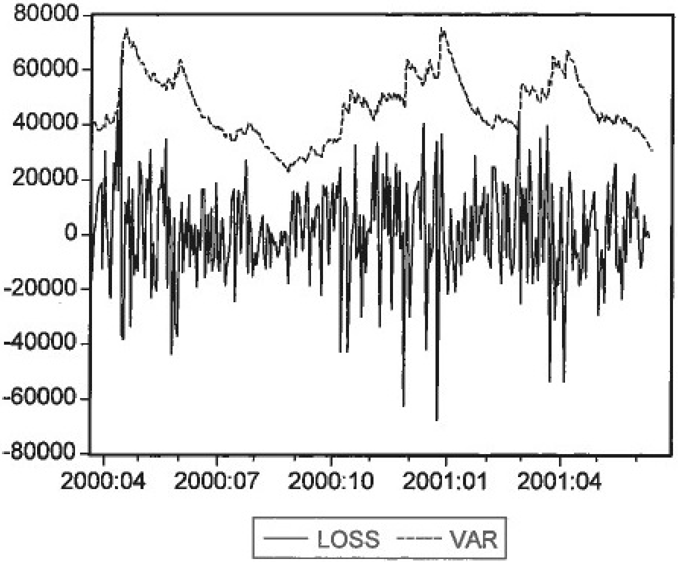
Portfolio



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Other Volatility Models: 1

- There are many variations on the GARCH model to address shortcomings and extend its applicability. We mention just a few of the more common ones.
- Consider GARCH-M (GARCH in the mean). Normally, we have a mean model and a separate volatility model.
- The GARCH-M model includes a volatility term in the mean equation. This can arise because it might be appropriate to include a risk premium associated with volatility in the returns part of the model

Other Volatility Models: 2

- This model can be expressed as

$$X_t = \mu + c\sigma_t^2 + a_t,$$

$$a_t = \epsilon_t \sigma_t,$$

$$\sigma_t^2 = \omega + \alpha_1 a_{t-1}^2 + \beta_1 \sigma_{t-1}^2.$$

- Here μ and c are constant. If $c > 0$ then the return process is positively related to the past volatility. Other specifications of the volatility term have been used including $c\sigma_t$ and $c\log(\sigma_t^2)$.

Other Volatility Models: 3

- One notable shortcoming of the GARCH model is the presence of volatility terms being squared $(a_{t-1}^2, \sigma_{t-1}^2)$
- This means that both positive and negative changes are treated the same in updating the volatility equation.
- However, one of the stylized facts is that the relationship should be asymmetric, with larger increases in volatility coming from negative changes than come from positive changes. the EGARCH (Exponential GARCH) model addresses this shortcoming.
- This was introduced in 1991 by Nelson. He considered weighting the innovations:

Other Volatility Models: 4

- Nelson began with the $\{\epsilon_t,\ t \geq 0\}$ innovations and weighted them $\{g(\epsilon_t),\ t \geq 0\}$ where

$$g(\epsilon_t) = \theta\epsilon_t + \gamma[|\epsilon_t| - E(|\epsilon_t|),$$

  where θ and γ are constants. Both components components are iid with mean 0. The asymmetry can be seen by writing out the two cases:

$$g(\epsilon_t) = \begin{cases} (\theta+\gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t \geq 0, \\ (\theta-\gamma)\epsilon_t - \gamma E(|\epsilon_t|) & \text{if } \epsilon_t < 0,. \end{cases}$$